MOHAMMED AMANSOUR

Electronics, Embedded Systems, Telecommunications

/mdamansour | /mdamansour | amansour.me

m@amansour.me - (+212) 708 99 28 70 - Fez, Morocco

TECHNICAL SKILLS

- **Programming:** Python, C/C++, Java, Assembly, LaTeX, VHDL
- **Software:** MATLAB, Simulink, PSIM, Multisim, Cadence Virtuoso, Quartus, ModelSim, VS code
- **OS:** Linux, Xenomai, Windows
- **Versioning:** Git, GitHub, GitLab
- **Hardware:** DSP, Arduino, STM32, Raspberry Pi, FOX G20, FPGA
- **Databases:** MySQL, phpMyAdmin
- **Web:** HTML, PHP, JS, CSS, XML, WordPress

LANGUAGES

- **Arabic:** Native
- **English:** Fluent
- **French:** Advanced
- **Spanish:** Conversational
- **German:** Beginner

CERTIFICATIONS

- **Institut Français:** DELF B2
- **EF SET:** English C2 (CEFR)
- **Huawei:** Next-Gen Cyber Security
- **APM Terminals:** HSSE
- **MathWorks:** MATLAB Onramp

VOLUNTEERING

- **Amansour Blog:** Education
- **Yallah Nette3awnou:** Food relief

SOFT SKILLS

Self-Management, Solution-Oriented, Problem-Solving, Reliability, Initiative, Critical Thinking, Communication

HOBBIES

Meeting International People, Simplifying complex concepts on YouTube and on my Blog, Chess, Running, Writing.

SUMMARY

M.Sc. candidate, and Embedded Systems Test Engineer at ALTEN. Through my current work in HIL system validation, I apply a strong foundation in MBD and hardware/software integration to test and verify complex embedded systems. I aim to leverage my expertise in model-based testing and test rig execution to ensure the rigorous safety and reliability of advanced flight control systems.

EDUCATION

M.Sc. in Electrical Engineering & Embedded Systems 2025–2027
Faculty of Science and Technology – USMBA, Fez
- Designed, simulated, and validated integrated circuits in *Cadence Virtuoso*.
- Built and verified *FPGA* digital systems in *VHDL* using *Quartus II* and *ModelSim*.
- Deployed a dual-kernel *RTOS* based on *Linux* and *Xenomai* using *Ipipe*.

B.Sc. in Electrical Engineering & Industrial Systems 2022–2025
Faculty of Science and Technology – UAE, Tangier
- Designed a traffic-light controller using *logic gates*, and *sequential logic*.
- Built and prototyped a sun-tracking system using *Arduino* and *light sensors*.

High School Diploma in Electrical Sciences & Technologies 2019–2022
Technical High School Moulay Youssef, Tangier
- Developed a traffic light control system using a *PIC microcontroller*.
- Analyzed and experimentally tested *operational-amplifier circuits*.

PROFESSIONAL EXPERIENCE

Embedded Test Engineer (Apprentice) | ALTEN, Fez Apr 2026 - Ongoing
- Developing and executing test strategies for automotive embedded systems.
- Developed a real-time suspension model for tractors with an interactive dashboard.
Tools: C, Python, Java, TCP/IP, Scrum, V-Cycle, CANoe, CANalayzer, HIL, GitLab.

Mobile App Developer (Co-Founder) | GryLab, Tangier Apr 2025 - Ongoing
- Developed and launched the first beta version of a workout-tracking application.
- Launched grylab.com and established the startup's core branding and operations.
Tools: Kotlin, Git, GitHub, Android Studio, WordPress, Google Play Console.

Embedded Systems Engineer (Intern) | APTIV, Tangier Apr–Jul 2025 (3 months)
- Designed and implemented a real-time embedded system on an MCU with HMI.
Tools: Proteus, Python, C++, I2C, Arduino Mega, Datasheets.

Automation Engineer (Intern) | SPIA Maroc, Ksar Sghir Sep 2024 (1 month)
- Configured an anti-collision safety system with real-time laser sensors and PLCs.
Tools: SOPAS ET, Simatic Step 7, MATLAB, Ladder Logic.

PERSONAL PROJECTS

- Developed an **ADAS** lane detection system in **MATLAB** using **Computer Vision**.
- Developed online e-commerce websites using **WordPress** and **WooCommerce**.
- Developed a **terminal-based (CLI)** chess game in **C**.